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MIL-DTL-5688B
15 August 1997
SUPERSEDING
MIL-C-5688A
16 May 1950

DETAIL SPECIFICATION

WIRE ROPE ASSEMBLIES; AIRCRAFT, PROOF TESTING AND PRESTRETCHING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification presents the standard method for the proof testing and prestretching of aircraft wire rope assemblies.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Supply Center, 700 Robbins Avenue, (DISC-EED), Philadelphia, PA 19111-5096 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4010

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SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-DTL-781 - Terminals; Wire Rope, Steel (For Swaging).
MIL-DTL-83420 - Wire Rope, Flexible, For Aircraft Control

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (NPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Proof/Prestretching load. Loads for proof load testing and prestretching of all wire rope terminal assemblies, including the entire length of the wire rope shall be as specified in table I. Proof load testing may also be used for the prestretching of the wire rope.

TABLE I. Load Values.

		Carbon Steel Wire Rope MIL-DTL-83420, Type I and II Composition A		Corrosion-Resistant Steel Cable MIL-DTL-83420, Type I and II Composition B	
Cable		Load (lbs)		Load (lbs)	
Diameter	Construction	Minimum	Maximum	Minimum	Maximum
1/32	3X7	66	69	66	69
3/64	7X7	162	170	162	170
1/16	7X7	288	302	288	302
1/16	7X19	288	302	288	302
3/32	7X7	552	580	552	580
3/32	7X19	600	630	552	580
1/8	7X19	1,200	1,260	1,056	1,109
5/32	7X19	1,680	1,764	1,440	1,512
3/16	7X19	2,520	2,646	2,220	2,331
7/32	7X19	3,360	3,528	3,000	3,150
1/4	7X19	4,200	4,410	3,840	4,032
9/32	7X19	4,800	5,040	4,680	4,914
5/16	7X19	5,880	6,174	5,400	5,670
3/8	7X19	8,640	9,072	7,200	7,560

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4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall be conducted on each assembly. Conformance inspection shall consist of load testing.

4.2 Load testing. The load shall be applied with the cable assembly lying in a straight line for the period of time specified as follows:

Type of Cable Assembly	Duration of Load Application (Min)
Swaged Terminal 1/	5 seconds

1/ Applies to swaged terminals conforming to MIL-DTL-781.

4.2.1 Application of load. The load shall be applied evenly by dead load or by any suitable test machine in not less than 3 seconds. The release shall be made evenly and gradually.

4.2.2 Rejection. Any cable terminal assemblies showing any slippage noticeable to the unaided eye shall be rejected. A suitable marking of the cable terminal joint shall be made prior to application of the load to aid in determination of any slippage.

4.2.3 Acceptance. Acceptance of the cable shall be indicated on a removable tag attached to the cable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification presents the standard method for the proof testing and prestretching of aircraft wire rope assemblies.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).

6.3 Subject term (key word) listing.

Wire rope
Terminals
Proof load
Swaged terminals

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MILITARY INTEREST

Custodians:
Army - AV
Navy - AS
Air Force - 11

Review activity:
Army - MI

Preparing activity:
DLA - IS

(Project 4010-0228)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-5688B	**2. DOCUMENT DATE (YYMMDD)** 970815

3. DOCUMENT TITLE WIRE ROPE ASSEMBLIES; AIRCRAFT, PROOF TESTING AND PRESTRETCHING OF

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME E. ALTOMARI	b. TELEPHONE *Include Area Code)* (1) Commercial (215) 697-6827 (2) AUTOVON 442-6827
c. ADDRESS *(Include Zip Code)* DEFENSE INDUSTRIAL SUPPLY CENTER, CODE DISC-EED, 700 ROBINS AVE, PHILADELPHIA, PA 19111-5096	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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